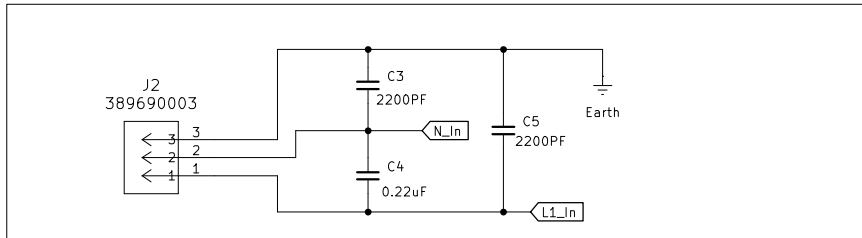
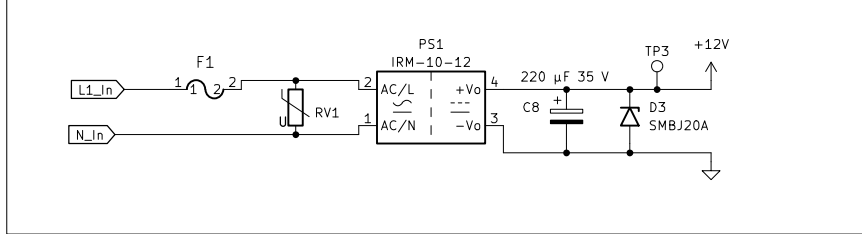


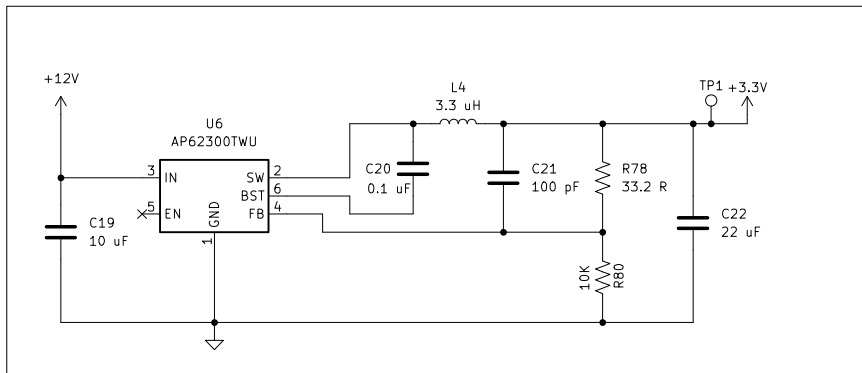
AC 1PH Input



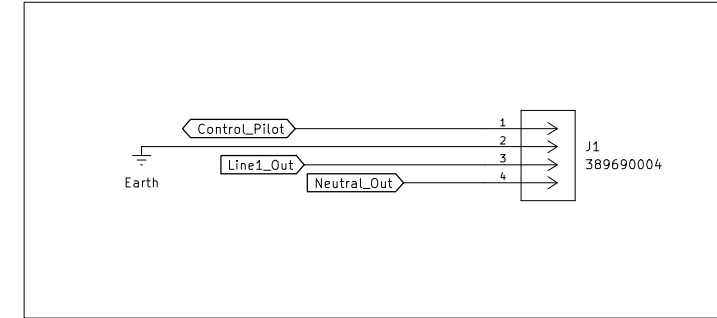
AC->DC converter



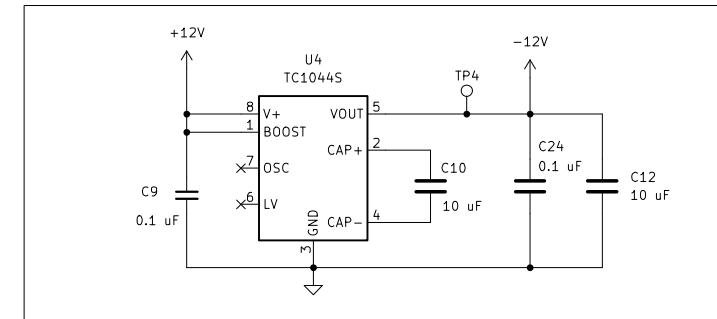
Power Supply MCU



Output Connectors



-12V Power regulator for CP-PP



Sheet: /Power Supply/
File: Power Supply.kicad_sch

Title:

Size: A4

Date:

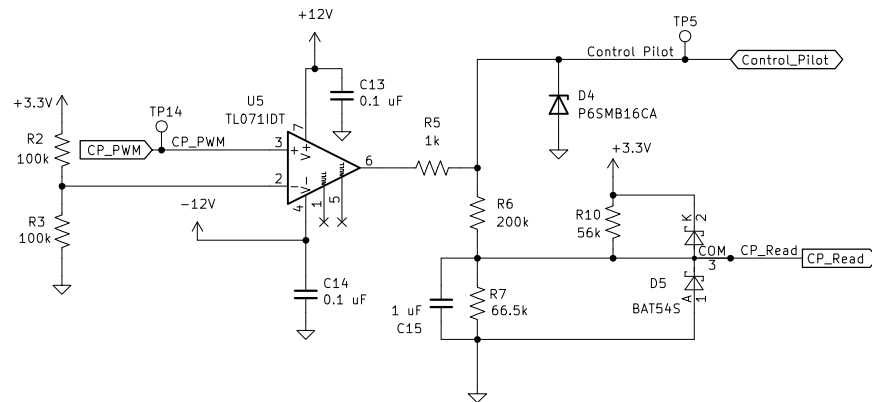
KiCad E.D.A. 8.0.4

Rev:

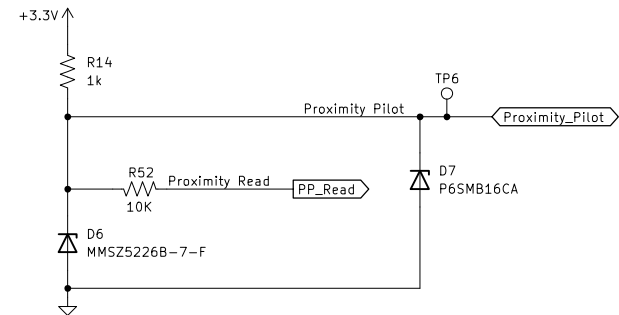
Id: 2/10

Control Pilot

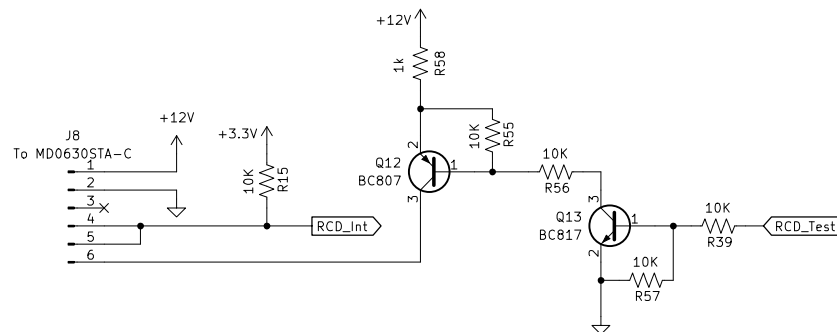
Alternate PN for CP: TLV1811DBVR from TI
Reference from
1. TI Implementing SAE J1772 Pilot Wire System in EV Charging Station With Comparators and
2. Microchip Ref design



Proximity Pilot



RCD Cable Type



RCD Test to verify the correct operation of the residual current sensing.
When the DC leakage current is detected to be 3-6mA or the AC leakage current is 15-30mA, DA1 will output an alarm signal (low level 0V).

Sheet: /RCD & CP Logic/
File: RCD_CP_Logic.kicad_sch

Title:

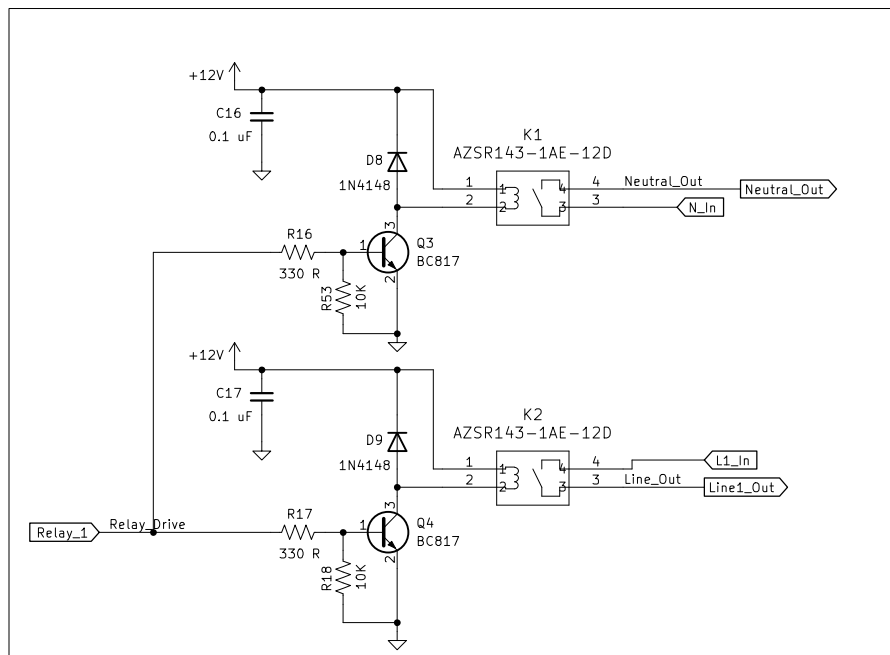
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KiCad E.D.A. 8.0.4

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Id: 3/10

Main Relays



Alternate PN for relay 2A HF116F-G1/12-2HTF

Sheet: /Relays/
File: Relays.kicad_sch

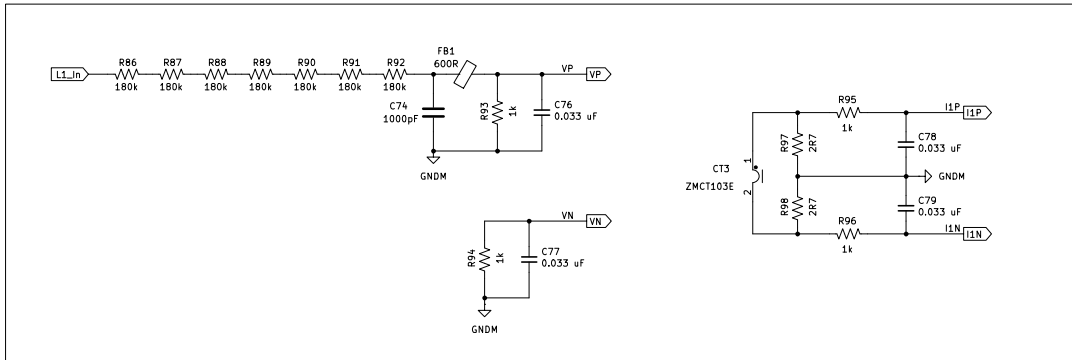
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KiCad E.D.A. 8.0.4

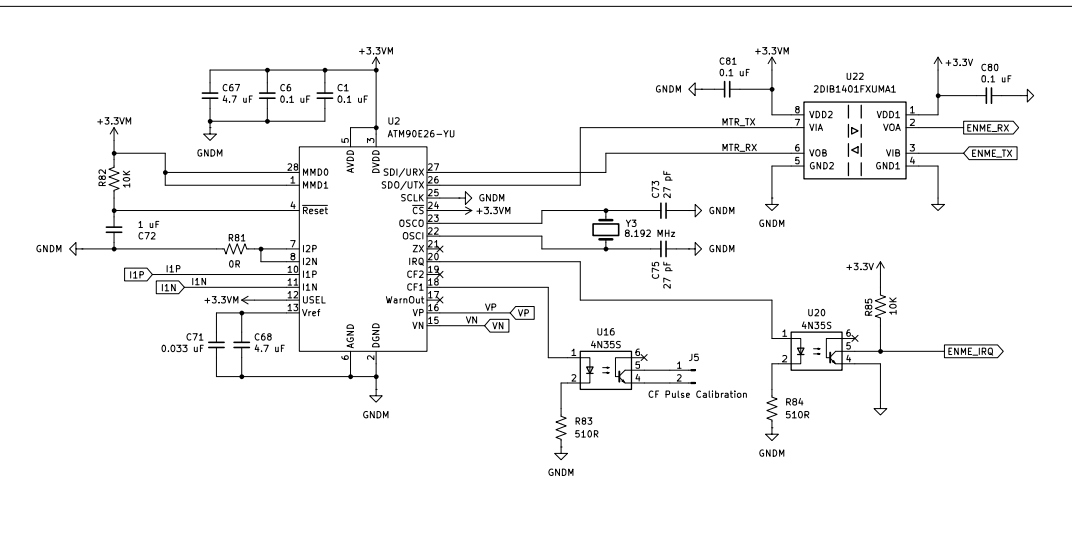
Date:

Rev:
Id: 4/10

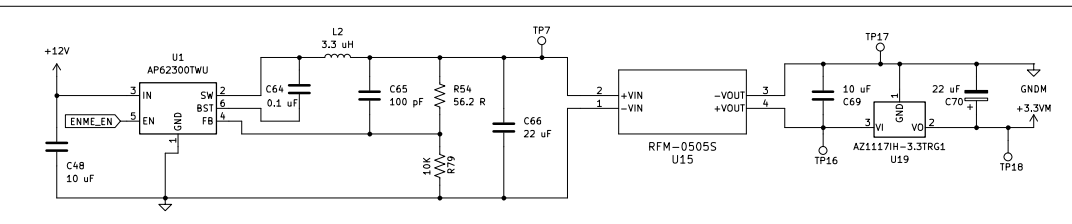
Voltage Divider and CT



Single Phase Metering



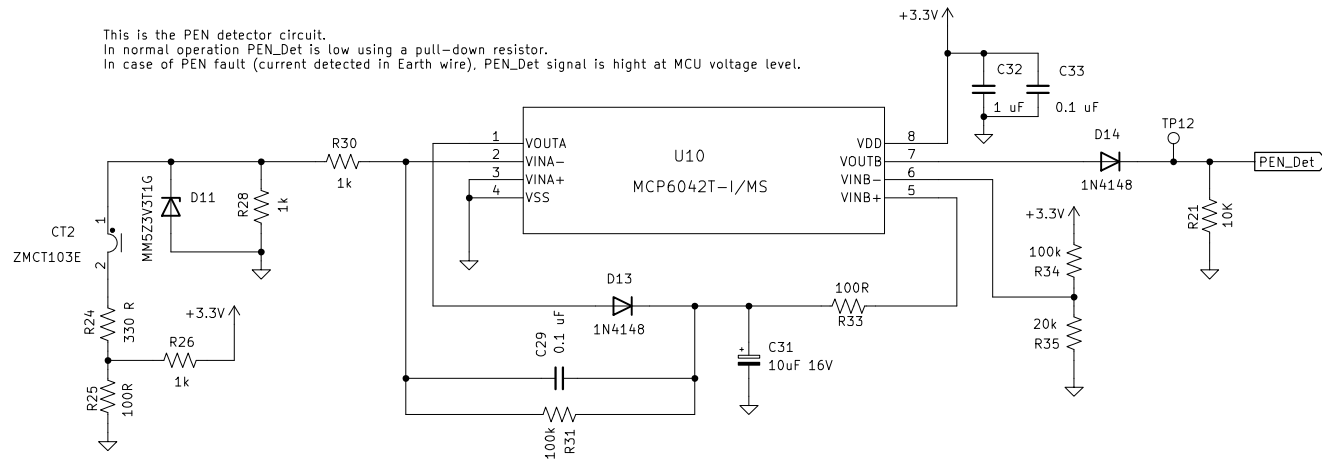
Power Supply for Metering



Sheet: /Energy Metering/		
File: Metering.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 8.0.4		Id: 5/10

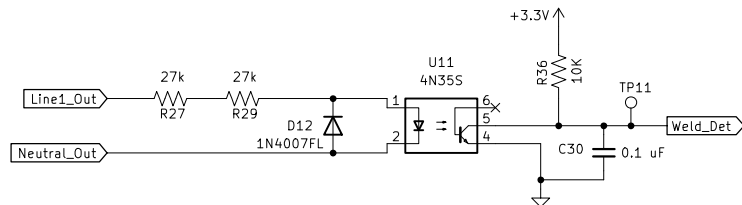
PEN Fault Detection

This is the PEN detector circuit.
In normal operation PEN_Det is low using a pull-down resistor.
In case of PEN fault (current detected in Earth wire), PEN_Det signal is high at MCU voltage level.



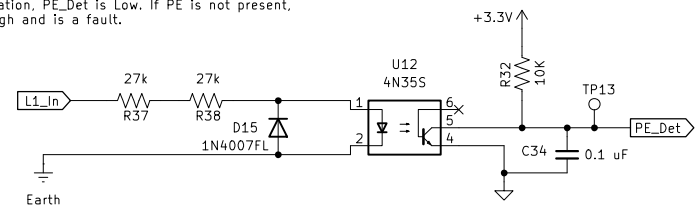
Weld Fault Detection

When the charger output is off, weld detect is HIGH.
Before starting and after stopping the charger it should test for Weld detect.



Earth Detection PE

Detect if Earth (PE) is connected and present at the output terminal.
If PE is floating or broken, this alerts the MCU via the PE_Det signal.
In normal operation, PE_Det is Low. If PE is not present, PE_det goes High and is a fault.



Sheet: /Safety & Fault Detection/
File: safety and Fault Detection.kicad_sch

Title:

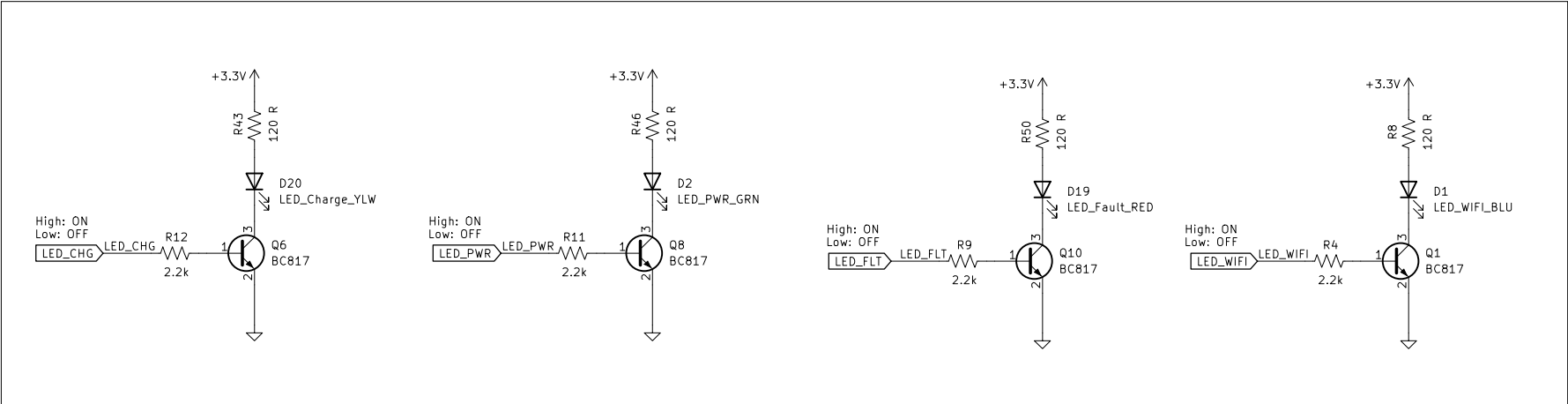
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KiCad E.D.A. 8.0.4

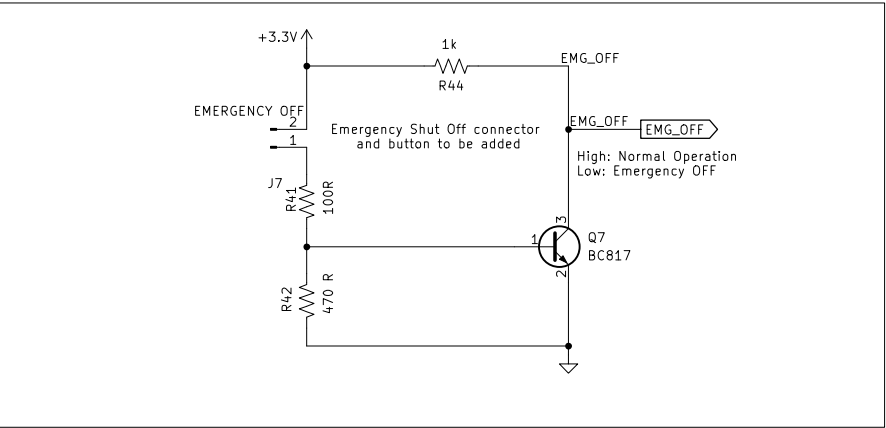
Rev:

Id: 6/10

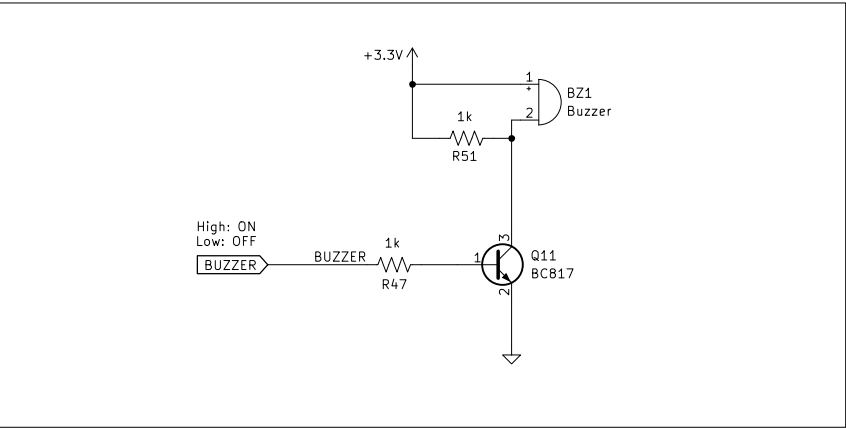
LED Indicators



Emergency Off Button



Fault Buzzer



Sheet: /Indicators LEDs/
File: Indicators_LED.kicad_sch

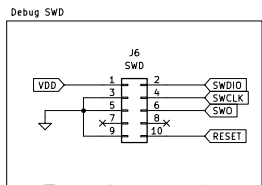
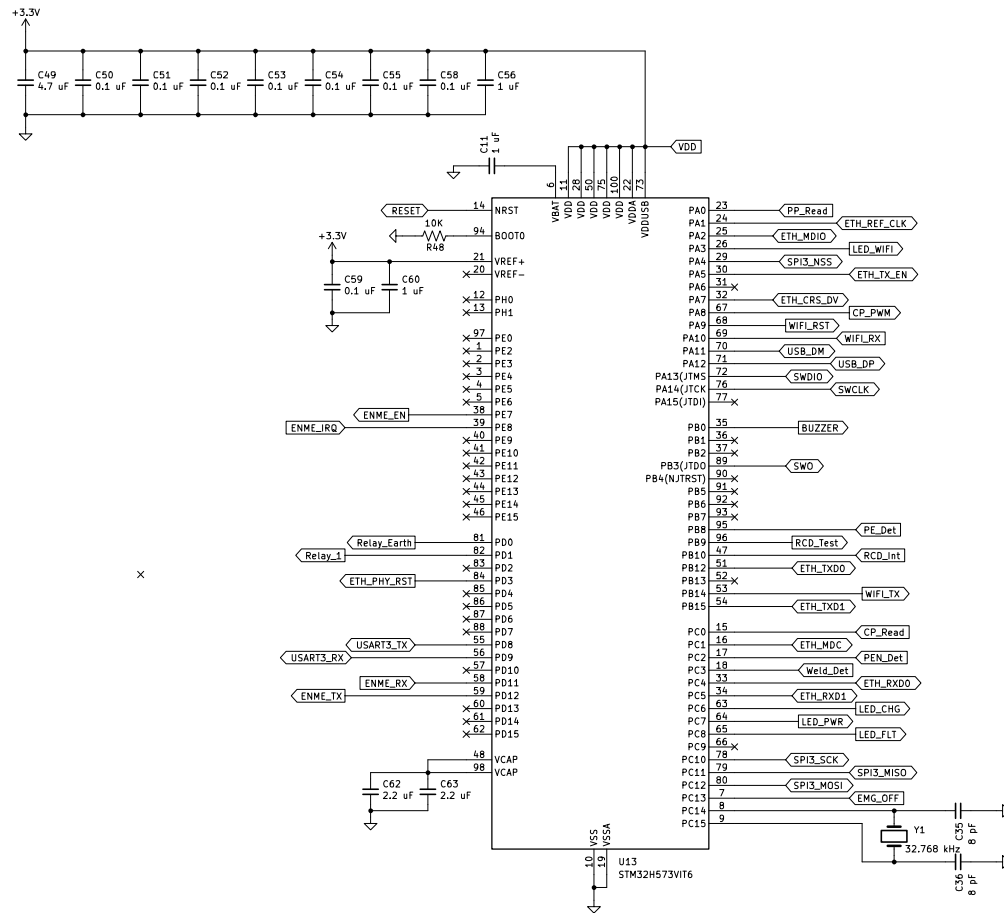
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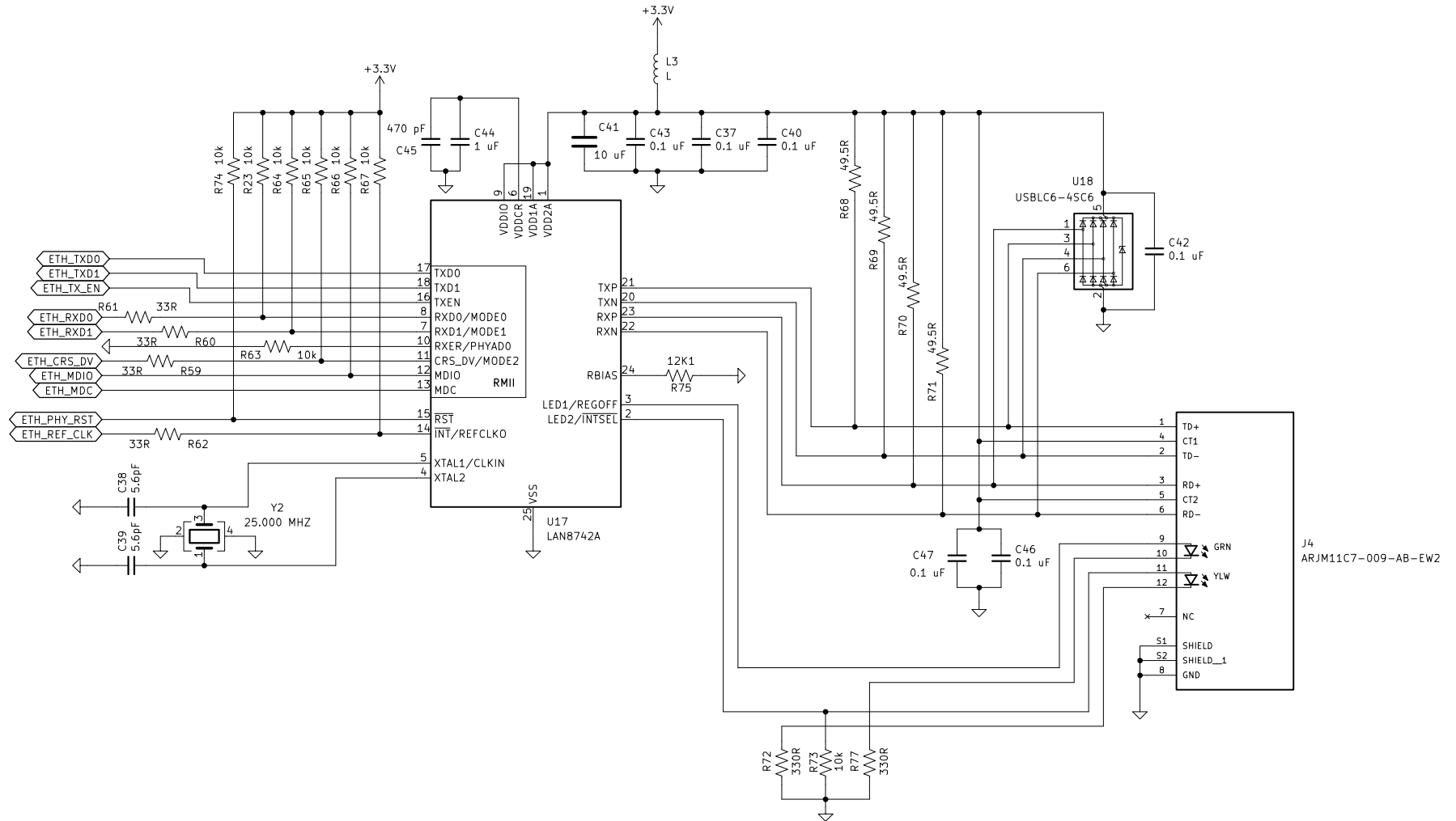
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Reviewed

Ethernet



Sheet: /Ethernet/
File: Ethernet.kicad_sch

Title:

Size: A4
KiCad E.D.A. 8.0.4

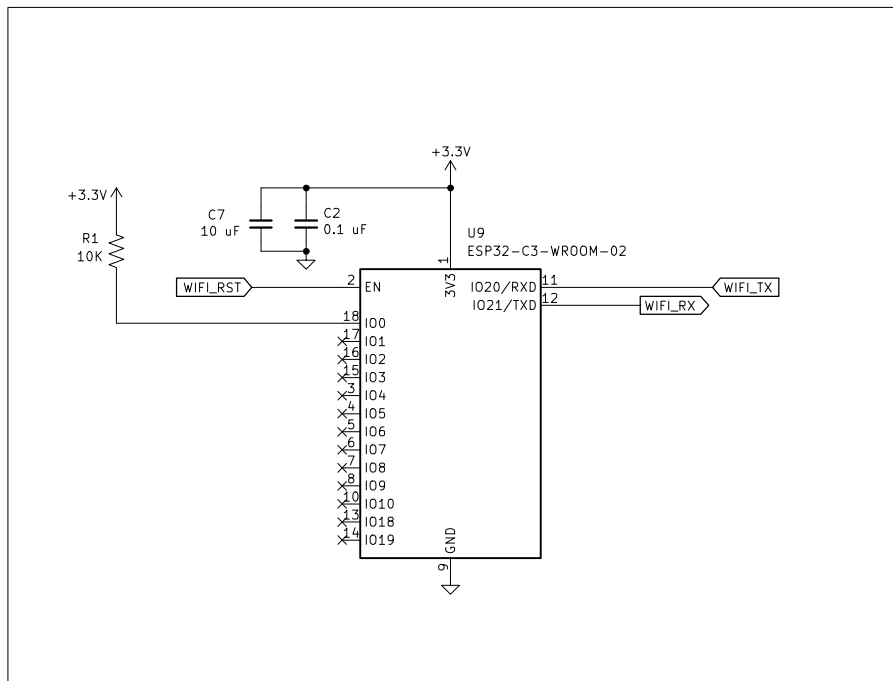
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Id: 9/10

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WiFi



Sheet: /WiFi/
File: WiFi.kicad_sch

Title:

Size: A4
KiCad E.D.A. 8.0.4

Date:

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Id: 11/10